20240617 EB_MN (DPR)

Project: Yi-Shuan Tseng Holt Lab
Author: Yi Shuan Tseng
Status: In progress
Entry Created On: 03 Jul 2024 18:17:10 UTC
Entry Last Modified: 23 Jul 2024 01:12:41 UTC
Export Generated On: 01 Sep 2026 18:29:51 UTC

2024年6月17日 星期一

split iPSCs into a full 6-well plate on the Friday before differentiation
differentiate 6 of 6-wells with 10mL T2 in cell-repellent plates

Table1

	A	B	C
1	genotype	passage	wells
2	WT-11	P34	6

2024年6月19日 星期三

T2 -> T3

2024年6月21日 星期五

T3 -> T4 (~14 mL for over-weekend)

2024年6月24日 星期一

T4 -> T5

2024年6月26日 星期三

T5 -> T6

2024年6月28日 星期五

T6 -> T6 (~14mL for over-weekend)

2024年7月1日 星期一

T6 -> T7
Dissociation

1. Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) for 4-5 hrs
2. Remove the PLO and wash the plate with PBS once
3. Coat the plate with laminine for 1hr.
4. Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs).
5. Add 3mL trypsin (usually 3-4 mL/10-cm plate) and pipette it up and down with 5mL serological pipette.
6. Put the tube into 37C water bath. Pipette the cells with P1000 every 5 min round 8-10 times until 15 min. (3 times in total. There might be more resistance at the 2nd round)

7. Add 6mL of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently. (Try to make the viscous cluster get through the tip to break it down)
8. Add 6mL of CTWM and pipette it with 5mL serological pipette gently.
9. Filter the cell with 40um filter into a new 50mL tube
10. Spin the cell down at 300rcf, 7min. Remove the supernatent.
11. Add 4mL T7 to resuspend the cells and count it.
12. Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

Table2

	A	B	C	D
1		conc (K/mL)	volume	total (M)
2	WT-11	795	4	3.18

add 400K cells into 8 wells and add 14uL of 110 or 111 to each 4 wells

Well1

	1	2	3	4
A	110		111	
B				
C				

2024年7月2日 星期二

Change T7 -> T7

2024年7月3日 星期三

T7-> BP T8

2024年7月5日 星期五

400uL BP T8 -> 500 uL BP T8

2024年7月8日 星期一

400uL BP T8 -> 500 uL BP T8

2024年7月10日 星期三

400uL BP T8 -> 500 uL BP T8

2024年7月12日 星期五

Imaging (DPR)

Table3

	A	B	C	D	E	F	G	H
1		stock (mM)	working (uM)	dilution	total media	drug added	total drug added	added media volume
2	PR30	10	10	1000	1600	1.6	1.76	225
3			1	10000	1600	0.16		take 20 uL from above and add 185 uL T8
4	GR30	10	10	1000	1600	1.6	1.76	225
5			1	10000	1600	0.16		take 20 uL from above and add 185 uL T8
6	GP30	1	1	1000	1600	1.6	1.6	200

Run every thing all together with ND sequencing acquisition
Setting: New spinning disk confocal, 600x900, 16bit, HDR, 15 min in total

O.C.

	A	B	C	D
1	channel	bit	power (%)	exposure time
2	640	16		
3	561	16		
4	488	16	50	1 s

plate map

	1	2	3	4
A	2	1	4	3
B	3	4	1	2
C				

font color: 110, 111
acquisition group: group1, group2
file name (cat) :

- pre-treatment: NT
- post-treatment: DPR

ND sequencing setting: **treat group1 -> wait for 12 min -> treat group 2 -> start at around 15min after group1 treatment.**

File ID map

	A	B	C	D	E	F	G
1		15-30 min	45-60 min	75-90min	105-120 min	135-150min	165-180min
2	group1	1,2	5,6	9,10	13,14	17, 18	21,22
3	group2	3,4	7,8	11,12	15,16	19, 20	23,24

position ID

	A	B	C	D
1	condition #		110	111
2	1	NT	1,2	1,2
3	2	PR30 1uM	3,4	3,4
4	3	GR30 1uM	5,6	5,6
5	4	GP30 1uM	7,8	7,8